

Param Jain

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EDUCATION

University of California San Diego, La Jolla, CA

June, 2028

BS in Microbiology: Expected to Graduate June 2028

Relevant Coursework: BILD 5 (Data Analysis/Design for Bio), BILD 4 (Introductory Biology Lab), BILD 1-3 (Foundational Biology), CHEM 6A-6B (Gen Chem I-II), BIPN 100 (Physiology), BICD 100 (Genetics)

Lynbrook High School, San Jose, CA

2025

Awards: Valedictorian

RESEARCH AND PROJECTS

First Author | Comparative Analysis of Deep Learning Models for Classifying Ulcerative

Colitis Stages on the Mayo Scale

October, 2024

Published in 2024 5th International Conference on Machine Learning and Computer Application (ICMLCA)

- Designed and implemented deep learning models to classify ulcerative colitis stages on the Mayo endoscopic scale using annotated medical imaging datasets.
- Compared the performance of multiple different models to optimize diagnostic accuracy.
- Applied statistical metrics to evaluate model performance and clinical reliability.

OraVise | Accessible Infrared Scanner for Dental Caries

September 2023 - Present

An Infrared scanner for at-home diagnosis of dental cavities

- Patent Pending (24ADI04P)
- Best in Show at the California Invention Convention, Accessibility Award at the National Invention Convention, Represented the USA at the AmericasX International Convention

Stanford Cardiothoracic Surgical Skills Summer Internship

Summer 2023

- Gained understanding of surgical methods, cardiothoracic anatomy, and physiology

Predicting Zoonotic Spillover Risk from Viral–Human Protein Interactions

Manuscript in Progress

EXTRACURRICULARS

Eagle Scout – Silver Palm

September 2018 – December 2025

- Organized campouts and advancement alongside fellow patrol leaders
- Experience in leading and teaching Cub Scouts about essential skills

We Love Stem Non-profit Chapter Director

June 2021 – June 2025

- Taught kids in 4th-8th grade the basics of biology, chemistry, and physiology
- 300+ service hours (Gold PVSA Award)

SKILLS

- *Laboratory:* Gel electrophoresis, PCR, DNA extraction, EcoPlate™ assays, bacterial culturing, Gram staining, aseptic technique, spectrophotometry, pipetting, and centrifugation.
- *Data & Analysis:* Experimental design, statistical testing (t-test, ANOVA, regression), data visualization, and scientific documentation.
- *Programming:* Intermediate R (data analysis, ggplot) and Java (object-oriented programming, data structures)